

CLOUD DATA AI ENGINEER

45-Day Intensive Upskill & Cross-Skill Roadmap

8–12 Hours Per Day / Non-Technical → Cloud Data AI Engineer

Fast-track, interview-focused preparation. The objective is practical competence, not mastery of every platform feature.

1. 45-Day Strategy

This 45-day version compresses the longer roadmap into an intensive preparation program. The learner should keep SQL and Python active every day, learn Hadoop primarily for concepts, and focus hands-on depth on PySpark, AWS, Databricks, Snowflake, dbt and Airflow.

- 8 hours/day = minimum intensive schedule.
- 10 hours/day = recommended standard.
- 12 hours/day = project/implementation days; do not make every day a 12-hour marathon.
- Target approximately 60% hands-on coding/building, 20% structured learning, 15% interview/architecture and 5% vocabulary/revision.
- Use AI coding agents from the first week as tutors and assistants, while reviewing all generated code.

2. Daily 10-Hour Template

Block	Hours	Activity
Morning	2	SQL / Python coding
Morning	1	Data Engineering vocabulary + revision
Late morning	2	Main technology concepts
Afternoon	2	Hands-on lab
Evening	2	Project implementation
Night	1	Interview questions / architecture / AI agent

3. 45-Day Roadmap at a Glance

Days	Focus	Outcome
Days 1–5	Foundations + Data Engineering Vocabulary + Linux	Understand the data lifecycle and core terminology
Days 6–12	SQL Intensive	Strong SQL and interview problem solving
Days 13–17	Python for Data Engineering	Practical Python, APIs, files, Pandas and testing
Days 18–22	Git + Agile + Big Data + Hadoop	Distributed-computing foundation
Days 23–29	PySpark + Spark Internals	Core large-scale processing skill
Days 30–34	AWS Data Engineering	S3, Glue, Athena, EMR, Redshift, Lambda and architecture
Days 35–37	Databricks + Delta Lake	Lakehouse and production Spark
Days 38–39	Snowflake + dbt	Cloud warehouse + transformation
Days 40–41	Airflow + Jenkins/CI-CD	Production orchestration and deployment
Days 42–43	AI Coding Agents + GenAI/RAG/Cortex	AI-augmented engineering
Days 44–45	Capstone + Architecture + Mock Interviews	Integration and job readiness

4. Detailed 45-Day Plan

Day	Focus	Topics	Deliverable
1	Data Engineering Landscape	Data Engineer vs Analyst/DS/ML Engineer; ingest → store → transform → serve → consume; OLTP vs OLAP.	Draw a complete data lifecycle.
2	Core Vocabulary I	ETL/ELT, batch/streaming, warehouse/lake/lakehouse/mart, structured/semi-structured/unstructured.	Create first 50 vocabulary cards.
3	Core Vocabulary II	Schema, table, partition, index, pipeline, orchestration, ingestion, transformation, data quality, lineage, governance.	Create 50 more vocabulary cards.
4	Core Vocabulary III	Idempotency, incremental/full load, CDC, SLA, latency, throughput, scalability, star/snowflake schema, facts/dimensions, SCD1/SCD2.	Explain each term aloud.
5	Linux + Environment	cd, ls, mkdir, cat, grep, pipes, permissions, ssh; install Python, Git and database tools.	Create project workspace and Git repo.
6	SQL Basics	SELECT, WHERE, ORDER BY, LIMIT, DISTINCT; aggregates.	20 SQL exercises.
7	SQL Grouping + NULLs	GROUP BY, HAVING, COUNT/SUM/AVG/MIN/MAX, CASE, COALESCE, NULL.	25 SQL exercises.
8	SQL Joins	INNER, LEFT, RIGHT, FULL, CROSS, SELF joins; cardinality.	25 join problems.
9	SQL CTE + Subqueries	WITH, correlated/non-correlated subqueries; set operations.	20 problems.
10	SQL Windows I	ROW_NUMBER, RANK, DENSE_RANK; PARTITION BY; ORDER BY.	25 window problems.
11	SQL Windows II	LAG, LEAD, running totals, moving averages, FIRST/LAST VALUE.	25 window problems.
12	SQL Advanced + Modeling	MERGE/UPSERT, DDL/DML, views, basic optimization, star schema.	Mini e-commerce SQL project.
13	Python Core I	Variables, types, strings, lists, tuples, sets, dictionaries, conditions, loops.	15–20 exercises.
14	Python Core II	Functions, comprehensions, lambda, exceptions, modules, packages, venv/pip.	15–20 exercises.
15	Python Data Engineering	CSV, JSON, text files, datetime, logging, environment variables, argparse.	Build file-processing utility.
16	Pandas + APIs	DataFrames, filter, groupby, merge, pivot, nulls; requests and JSON APIs.	API → Pandas transformation.
17	Python DB + Testing	PostgreSQL/SQLAlchemy concepts, transactions, pytest, clean modular functions.	API → Python → PostgreSQL ETL.
18	Git + Agile	init, clone, add, commit, push/pull, branches, merge, conflicts; Scrum, sprints, stories.	Create PR-style workflow.
19	Big Data Concepts	5 Vs, distributed computing, horizontal/vertical scaling, locality, replication, fault tolerance.	Explain why distributed systems are needed.
20	HDFS	NameNode, DataNodes, blocks, replication, rack awareness, read/write flow, hdfs dfs commands.	HDFS architecture lab.
21	MapReduce + YARN	Map → Shuffle/Sort → Reduce; ResourceManager, NodeManager,	WordCount + YARN architecture.

		ApplicationMaster, containers.	
22	Sqoop + Hive	Sqoop import/export, split-by, incremental concepts; HiveQL, managed/external tables, partitions, buckets, Metastore, ORC/Parquet.	Partitioned Hive mini-project.
23	Spark Fundamentals	Driver, Executors, Cluster Manager, SparkSession; RDD/DataFrame; lazy evaluation.	Draw Spark architecture.
24	Spark Execution	Application → Job → Stage → Task; DAG; transformations vs actions; client vs cluster.	Trace an example Spark job.
25	PySpark DataFrame I	Read/write CSV, JSON, Parquet; select, filter, withColumn, cast, when/otherwise.	10 DataFrame exercises.
26	PySpark DataFrame II	groupBy, agg, joins, Spark SQL, temp views, window functions.	15 PySpark exercises.
27	Spark Performance I	Partitions, repartition/coalesce, shuffle, broadcast joins.	Optimize join exercises.
28	Spark Performance II	Skew, salting, caching, Catalyst, Tungsten, AQE, Spark UI.	Diagnose a slow job.
29	PySpark Project	Multi-GB dataset: clean, join, aggregate, window, partitioned Parquet, optimization.	Complete Spark project.
30	AWS Foundations	IAM, roles/policies, regions/AZs, EC2/VPC basics, S3 buckets/prefixes/storage classes.	Create secure S3 data-lake layout.
31	AWS Glue + Catalog	Glue crawlers, Data Catalog, Glue ETL, Glue Studio; Glue PySpark.	Build S3 → Glue → Parquet.
32	Athena + EMR + Redshift	Athena serverless SQL, partitions/cost; EMR vs Glue; Redshift warehouse concepts.	Build/query a cloud pipeline.
33	AWS Event/Streaming Services	Lambda, Kinesis Data Streams/Firehose, Step Functions, DMS, RDS/DynamoDB, SNS/SQS, CloudWatch.	Design batch + streaming architectures.
34	AWS End-to-End Project	S3 raw → Glue Catalog → Glue PySpark → S3 silver/curated → Athena/Redshift; IAM and monitoring.	Complete AWS pipeline.
35	Databricks Fundamentals	Workspace, notebooks, clusters/job clusters, Runtime, Workflows.	Build notebook-based pipeline.
36	Delta Lake	ACID, transaction log, time travel, MERGE, UPDATE/DELETE, OPTIMIZE, Z-ORDER, VACUUM.	Implement Delta CDC/SCD-style flow.
37	Databricks Architecture	Bronze → Silver → Gold; Auto Loader; Unity Catalog; data quality.	Complete lakehouse mini-project.
38	Snowflake	Storage/compute separation, virtual warehouses, micro-partitions, stages, file formats, COPY INTO, Snowpipe, RBAC.	Build Snowflake loading demo.
39	dbt	Models, sources, ref/source, Jinja, materializations, tests, snapshots, seeds, macros, documentation and lineage.	Build staging → intermediate → marts.
40	Airflow	DAGs, tasks, operators, sensors, hooks, scheduling, dependencies, XComs, variables, connections, retries, backfill, catchup, idempotency.	Build ingest → transform → dbt DAG.
41	Jenkins + CI/CD	Jenkinsfile, pipeline stages, GitHub webhook, GitHub Actions concepts; SQL/Python/dbt/DAG validation.	Create CI pipeline.
42	AI Coding Agents	GitHub Copilot, Claude Code, Codex, Cursor, Windsurf, Cortex/Cortex Code; prompting,	Use 2–3 agents on the same repo.

		context, review, testing, refactoring.	
43	GenAI + RAG	LLMs, tokens, context windows, embeddings, vector DBs, RAG, agents, tool calling, data quality for AI.	Build a small RAG proof of concept.
44	Capstone Build	Integrate source → S3 → PySpark/Databricks → Snowflake → dbt → Airflow → CI/CD → AI.	Finish architecture, code and README.
45	Final Assessment	SQL + Python + PySpark + AWS + Databricks + Snowflake + dbt + Airflow + architecture + behavioral.	Mock interview + gap analysis + 30-day follow-up plan.

5. End-to-End Cloud Data AI Architecture

The learner should be able to explain and implement the following integrated flow:

1. Source systems: APIs, RDBMS, files and SaaS applications.
2. Ingestion: AWS Glue/DMS/APIs or equivalent ingestion mechanisms.
3. Data lake: S3 raw/bronze.
4. Processing: PySpark on Glue and/or Databricks.
5. Lakehouse: Delta Bronze → Silver → Gold.
6. Warehouse: Snowflake.
7. Transformation: dbt staging → intermediate → marts.
8. Orchestration: Airflow.
9. CI/CD: GitHub + Jenkins or GitHub Actions.
10. AI layer: Snowflake Cortex and/or RAG application.
11. Consumers: BI, analytics, ML and AI applications.

6. Capstone Project

Recommended final project: AI-Powered Customer/Sales Data Platform.

- Sources: API + relational database + CSV/JSON files.
- Landing: S3 Bronze.
- Processing: AWS Glue or Databricks PySpark.
- Lakehouse: Delta Silver/Gold.
- Warehouse: Snowflake.
- Transformation: dbt with incremental models, tests and SCD Type 2.
- Orchestration: Airflow.
- CI/CD: GitHub + Jenkins/GitHub Actions.
- AI: Cortex summarization/classification or a simple RAG interface.
- Documentation: architecture diagram, README, data dictionary, operational notes and interview explanation.

7. Portfolio Deliverables

#	Project	Primary Skills
1	SQL E-commerce Analytics	Advanced SQL + dimensional modeling
2	Python API → PostgreSQL ETL	Python + Pandas + APIs + testing
3	Hadoop/Hive Mini Pipeline	HDFS + MapReduce + YARN + Hive
4	PySpark Large-Scale Processing	Spark + performance optimization
5	AWS Data Lake	S3 + Glue + Athena + PySpark
6	Databricks Lakehouse	Delta + Medallion + Unity Catalog
7	Snowflake + dbt	Warehouse + incremental + tests + SCD2
8	Airflow/CI Pipeline	DAG + dbt + CI/CD
Final	Cloud Data AI Platform	Full-stack capstone

8. Daily Interview Preparation

- SQL: 15–20 problems/exercises daily.
- Python: 5–10 coding exercises daily.
- Vocabulary: 10 terms daily.
- From Day 23: 2–5 PySpark exercises daily.
- From Day 30: one AWS/architecture question daily.
- From Day 38: one Snowflake/dbt design or troubleshooting question daily.
- From Day 40: one Airflow/CI-CD question daily.
- Every evening: explain one concept aloud for 5 minutes.

9. 45-Day Target Metrics

Area	45-Day Target
SQL	500+ problems/exercises
Python	200+ exercises
PySpark	100+ exercises
Vocabulary	250–300 terms
Hadoop/Hive	Concept mastery + 1 lab
AWS	20+ hands-on labs
Databricks	10+ labs
Snowflake	10+ labs
dbt	20+ models
Airflow	8–10 DAGs
Git/CI-CD	10+ workflows
AI Agents	20+ practical coding tasks
Architecture	20+ design/explanation exercises
Major Projects	5–8 progressive projects
Capstone	1 integrated production-style project

10. Priority Levels

- CRITICAL: SQL, Python, PySpark, AWS, Data Engineering concepts and Git.
- HIGH: Databricks, Delta Lake, Snowflake, dbt, Airflow, CI/CD and AI coding agents.
- MEDIUM: Hadoop, Hive, MapReduce, YARN, Jenkins and Agile.
- CONCEPT-FOCUSED: Sqoop and older Hadoop ecosystem components.
- GROWING EDGE: GenAI, RAG, vector concepts and Snowflake Cortex.

11. Final Readiness Checklist

- ☐ Can I solve medium SQL problems using joins and window functions without copying solutions?
- ☐ Can I write a reliable Python ETL with logging, error handling and tests?
- ☐ Can I explain HDFS, MapReduce and YARN?
- ☐ Can I explain Spark Driver, Executor, Job, Stage and Task?
- ☐ Can I diagnose shuffle, skew and partitioning problems?
- ☐ Can I design an S3-based AWS data lake?
- ☐ Can I build a Glue PySpark pipeline?
- ☐ Can I explain Delta Lake, MERGE and CDC?
- ☐ Can I explain Snowflake storage/compute and micro-partitions?
- ☐ Can I build dbt incremental models and tests?
- ☐ Can I build an idempotent Airflow DAG?
- ☐ Can I explain Git and CI/CD for data engineering?
- ☐ Can I safely use Claude Code/Codex/Cursor/Copilot to accelerate engineering?
- ☐ Can I explain RAG, embeddings and vector databases?
- ☐ Can I design the complete Cloud Data AI platform?
- ☐ Can I explain my capstone architecture and engineering decisions?

12. Final Principle

One business problem → many technologies.

Do not learn AWS, Databricks, Snowflake, dbt and Airflow as isolated courses. Take one realistic data problem and progressively implement it with SQL, Python, Hadoop/Hive, PySpark, AWS, Databricks, Snowflake, dbt, Airflow, CI/CD and AI.

The 45-day objective is interview-ready practical competence. Deep production mastery will continue after the 45 days through real projects and employment.

Cloud Data AI Engineer — 45-Day Intensive Roadmap

13. 15 Core Data Engineering Skills for 2026

The following 15-skill framework is added to the 45-day roadmap as a parallel competency checklist. It complements the technology sequence by focusing on the engineering capabilities that keep a Data Engineer relevant.

#	Core Skill	Key Topics
1	1. SQL & Query Optimization	Joins, window functions, CTEs, execution plans
2	2. Data Modeling	Facts, dimensions, star schema, grain, normalization
3	3. Batch Processing	ETL, ELT, scheduling, backfills, idempotency
4	4. Streaming Systems	Kafka, Pulsar, event streams, windowing, consumers
5	5. CDC	WAL, Debezium, change capture, sync pipelines
6	6. Data Warehousing	Snowflake, BigQuery, Redshift, warehouse design
7	7. Lakehouse Fundamentals	Delta Lake, Iceberg, Hudi, ACID tables
8	8. Distributed Processing	Spark, partitions, shuffles, skew, optimization
9	9. Orchestration	Airflow, Dagster, retries, dependencies, SLAs
10	10. File Formats & Storage	Parquet, Avro, ORC, object storage, partitioning
11	11. Data Quality & Testing	Freshness, null checks, schema tests, dbt tests
12	12. Observability	Logs, metrics, lineage, alerts, pipeline monitoring
13	13. Cloud & Infrastructure	AWS, GCP, Azure, IAM, compute, storage
14	14. Python for Data Engineering	APIs, automation, transformations, validation
15	15. Cost & Performance Optimization	Pruning, clustering, caching, workload tuning

14. How the 15 Skills Map to the 45-Day Roadmap

Days	Roadmap Phase	15-Skill Competencies
Days 1–5	Data Engineering Foundations	Data Modeling; Batch Processing; File Formats & Storage; Cloud & Infrastructure
Days 6–12	SQL Intensive	SQL & Query Optimization; Data Modeling
Days 13–17	Python for Data Engineering	Python for Data Engineering; Data Quality & Testing
Days 18–22	Big Data + Hadoop	Distributed Processing; File Formats & Storage; Batch Processing
Days 23–29	PySpark + Spark Internals	Distributed Processing; Cost & Performance Optimization
Days 30–34	AWS Data Engineering	Cloud & Infrastructure; Batch Processing; File Formats & Storage; Observability
Days 35–37	Databricks + Delta Lake	Lakehouse Fundamentals; Distributed Processing; Data Quality & Testing
Days 38–39	Snowflake + dbt	Data Warehousing; Data Quality & Testing; Cost & Performance Optimization
Days 40–41	Airflow + CI/CD	Orchestration; Observability; Batch Processing; Data Quality & Testing
Days 42–43	AI Agents + GenAI	Python for Data Engineering; Observability; Data Quality & Testing
Days 44–45	Capstone + Architecture	All 15 skills integrated into one production-style platform

15. 15-Skill Interview Checklist

By the end of Day 45, the learner should be able to explain the following without relying on memorized definitions:

- How do joins, window functions and CTEs affect SQL performance? How do you read an execution plan?

- ☐ How do you define the grain of a fact table and design a star schema?
- ☐ How do you design an idempotent batch pipeline with scheduling and backfills?
- ☐ How would you design a Kafka/Pulsar streaming pipeline with consumers and windowing?
- ☐ How does CDC work using WAL/Debezium and how do you make synchronization reliable?
- ☐ When would you choose Snowflake, BigQuery or Redshift and how would you design the warehouse?
- ☐ What problem does a lakehouse solve, and how do Delta Lake, Iceberg and Hudi provide table-management capabilities?
- ☐ What causes Spark partitions, shuffles and skew, and how do you optimize them?
- ☐ How do you design an Airflow DAG with retries, dependencies, SLAs and idempotent tasks?
- ☐ When should you use Parquet, Avro or ORC, and how does object-storage partitioning affect performance?
- ☐ How do you implement freshness, null, schema and dbt tests?
- ☐ How do you monitor a production data pipeline using logs, metrics, lineage and alerts?
- ☐ How do IAM, compute and object storage fit into a cloud data platform across AWS/GCP/Azure?
- ☐ How do you build Python automation, transformations and validation for data pipelines?
- ☐ How do pruning, clustering, caching and workload tuning reduce data-platform cost and improve performance?

16. Updated 45-Day Success Model

The 45-day roadmap now has two parallel tracks: (1) the technology learning sequence and (2) the 15 core engineering competencies. Every project should be evaluated against both tracks.

- Technology track: SQL → Python → Hadoop → PySpark → AWS → Databricks → Snowflake → dbt → Airflow → CI/CD → AI.
- Engineering track: SQL optimization → modeling → batch → streaming → CDC → warehousing → lakehouse → distributed processing → orchestration → storage → quality → observability → cloud → Python → cost/performance.
- Final capstone requirement: demonstrate as many of the 15 competencies as are relevant to the chosen business problem, rather than merely showing a collection of disconnected tools.

Updated: 45-Day Roadmap + 15 Core Data Engineering Skills